Bitcoin Price Feature Importance Analysis

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Research Background

- Bitcoin prices are highly volatile and influenced by multiple factors.
- Machine learning methods can be used to predict Bitcoin prices.
- This study employs a Random Forest model to analyze the impact of different features on Bitcoin prices.

Data Collection and Preprocessing

- Data sources: SP 500, CSI 300, Brent Oil, Gold Futures, etc.
- Processing steps:
  1. Standardizing date formats.
  2. Selecting price data.
  3. Merging datasets.
  4. Handling missing values.

Feature Engineering

- Processing date information:
  - Extracting **day** from date.
  - Applying cyclic encoding (sin and cos transformations).
- Selecting other features: stock indices, commodity prices, etc.

Feature Correlation

- Calculating the **Spearman correlation coefficient** between features and Bitcoin price.
- Visualizing the correlation heatmap.

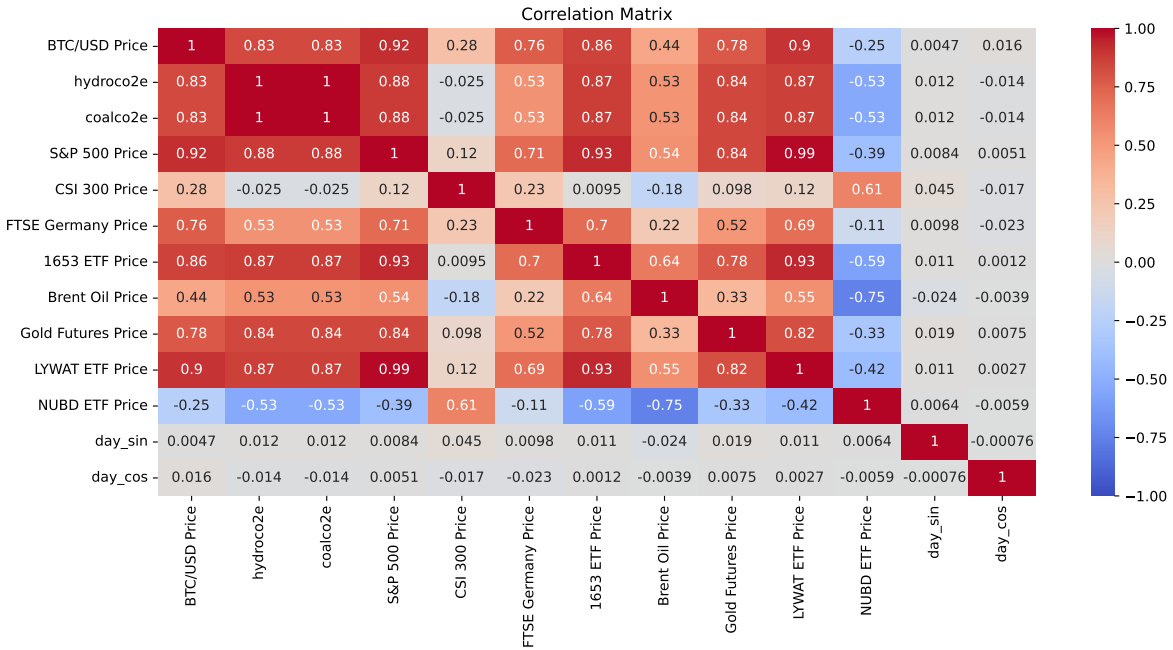


Figure: Feature Correlation Heatmap

Random Forest Model

- Target variable: Bitcoin price.
- Data split: 70% training, 30% testing.
- Using **RandomizedSearchCV** for hyperparameter tuning.

Best Hyperparameters

- n_estimators: 100
- max_depth: 10
- min_samples_split: 2
- min_samples_leaf: 1

Feature Importance

- Calculating and ranking feature importance.
- Visualizing feature impact.

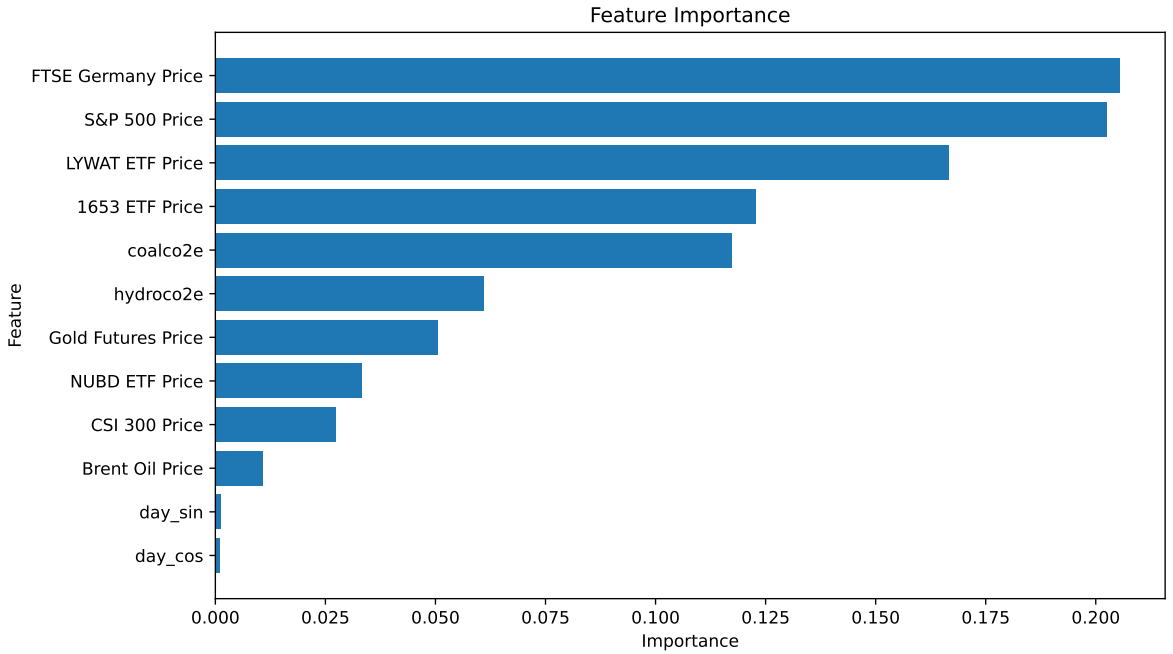


Figure: Feature Importance Ranking

Study Summary

- Successfully trained a Random Forest model to predict Bitcoin prices.
- Identified key influencing factors such as gold, oil, and SP 500.
- Future work may explore LSTM, XGBoost for further optimization.